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Review article



Effects of video game intervention on attention deficit in children with ADHD: A systematic review and meta-analysis

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BRIEF COMMUNICATIONS

Purpose: To explore the effects of a video game intervention on children with attention deficit hyperactivity disorder (ADHD).
Methods: Randomized controlled trials (RCTs) were systematically searched in PubMed, Web of Science, ScienceDirect, and X-MOL databases. Studies were screened based on predefined inclusion and exclusion criteria. The methodological quality of the included studies was assessed using the Cochrane Risk of Bias Assessment Tool. Relevant data were analyzed using RevMan 5.3.5 software, and publication bias was assessed using Stata 17.0.
Results: A total of 10 RCTs involving 820 subjects were included. Meta-analysis indicated that the video game intervention had a positive effect on reducing attention deficit symptoms in children with ADHD (SMD=-0.19, 95 % CI [-0.34, -0.03]). Egger's test confirmed that the included studies were free from publication bias.
Conclusion: Findings indicate that video game interventions show statistically significant but not clinically meaningful benefits on attention and may be therapeutic when added to other evidence-based therapies.

1. Introduction

1.1. ADHD

Inattention, one of the core symptoms of children with attention deficits hyperactivity disorder (ADHD), is a primary cause of these negative effects (Bental and Tirosh, 2007; Greven et al., 2014; Skalski et al., 2021; Trane and Willcutt, 2023), making it one of the most pressing issues for these children. Currently, there are both pharmacologic and non-pharmacologic approaches to the treatment of ADHD (S. Cortese et al., 2018; Mechler et al., 2022; Fridman et al., 2015). But there are some side effects of such drugs, such as insomnia, dry mouth, loss of appetite, and syncope (S. Cortese et al., 2018), and even affect children's growth and development (Greenhill et al., 2020). With the continuous advancements in this field, more pharmacological treatments have emerged, such as polyunsaturated fatty acids (Gillies et al., 2023). Non-pharmacological approaches mainly involve psychosocial interventions (Zwi et al., 2011), including cognitive-behavioral therapy, family therapy, and interpersonal psychotherapy (Correll et al., 2021). Other non-pharmacological treatments include neurofeedback (Cortese et al., 2016), video game therapy (Jaklevic, 2020), and brain stimulation (Rubia et al., 2021).

1.2. Video games and ADHD

Inattention in children with ADHD is primarily due to abnormal central nervous system function, characterized by poor inhibition and inadequate secretion of dopamine and norepinephrine due to dysfunction in the frontal-striatal circuitry (Christiansen et al., 2019). This dysfunction leads to attentional selection and maintenance issues when external stimuli are present. Studies have shown that video games can promote striatal dopamine release, which may improve attention deficit symptoms (Koepp et al., 1998).

Video games as an emerging digital therapy method, are increasingly used to diagnose and intervene in clinical symptoms in children with ADHD (Peñuelas-Calvo et al., 2022). In the clinical diagnosis of ADHD,

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virtual reality video game models have been employed to assess children's attention, impulsivity, and processing speed, effectively distinguishing children with ADHD from controls (Areces et al., 2018). Additionally, video games have been utilized to differentiate between ADHD subtypes, revealing a positive correlation between reaction time measured by the video game and inattentiveness subtypes (Delgado-Gomez et al., 2017). Studies indicate that video games have a high accuracy rate in diagnosing ADHD symptoms, with a 78 % accuracy rate for the inattentive subtype and a 75 % accuracy rate for the combined subtype of ADHD (Heller et al., 2013).

Interventions using video games for ADHD symptoms primarily focus on cognitive training and neurofeedback. One study conducted a 10-week, thrice-weekly PC video game intervention for children with ADHD, resulting in improvements in life skills and a reduction in ADHD symptoms (Bul et al., 2016). Attention deficit, a core symptom of ADHD, has been the primary target of video game interventions due to their potential ameliorative effects. However, with increased research, some studies have found that while video games may improve impulsivity, they do not significantly enhance attention in ADHD patients (Bul et al., 2018). Furthermore, some studies are limited by sample size, making their results less reliable.

This paper reviews related studies and employs meta-analysis to investigate whether video games have a significant effect on attention in children with ADHD.

2. Methods

This study adheres to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Liberati et al., 2009) for selecting and employing research methods. The review protocol was registered in the PROSPERO database (registration number CRD42024554539).

2.1. Search strategy

A systematic literature search was performed in four databases: PubMed, Web of Science, ScienceDirect, X-MOL. The final search date was May 10, 2024. The following search terms were used: ("attention deficit disorder with hyperactivity"[MeSH Terms] OR ("attention"[All Fields] AND "deficit"[All Fields] AND "disorder"[All Fields] AND "hyperactivity"[All Fields]) OR "attention deficit disorder with hyperactivity"[All Fields] OR "adhd"[All Fields]) AND ("child"[MeSH Terms] OR "child"[All Fields] OR "children"[All Fields] OR "child s"[All Fields] OR "child s"[All Fields] OR "children's"[All Fields] OR "children"[All Fields]) AND ("video games"[MeSH Terms] OR ("video"[All Fields] AND "games"[All Fields]) OR "video games"[All Fields] OR ("video"[All Fields] AND "games"[All Fields]) OR "video games"[All Fields]). Additionally, references from identified articles were screened for inclusion.

2.2. Inclusion and exclusion criteria

Subjects were patients aged 12 years and younger with a clinical diagnosis of ADHD; (2) implemented video games-based interventions; (3) reported the effects of video games on ADHD symptoms of attention deficit in children; (4) were randomized controlled trials (RCTs); (5) were published in peer-journals.

Studies were excluded if they met the following criteria: (1) included participants with other types of disabilities or failed to provide data specific to children with ADHD; (2) did not involve video games interventions; (3) were observational in nature (e.g., cross-sectional, case-control, or cohort); (4) were review articles, case/government reports, conference papers, book chapters, or policy documents.

2.3. Study selection

The literature was examined to ensure consistency with the research questions and to meet sufficient methodological quality standards. The included studies were assessed for risk of bias using the Cochrane Risk of Bias Tool. Two authors (PM and ZX) independently assessed the included studies, and any inconsistencies were resolved by involving the third (KY) and fourth (PZ) authors. Intraclass correlation coefficients were used to measure consistency.

2.4. Data extraction

Data were identified, examined, and mined by two independent authors (PM and ZX), after which both authors received systematic training and developed a data extraction form. Variables collected included authors, country, year of publication, design, age of sample, sample size, gender distribution of the sample, mean, standard deviation, and measures. The primary outcome was the effectiveness of the video game intervention on attention in children with ADHD.

2.5. Risk of bias assessment

Adoption of the Cochrane quality assessment handbook (Higgins et al., 2011). Risk of bias assessment of the literature was used to identify random sequence generation, allocation concealment, blinding, incomplete data, selective reporting and other biases. The process was completed independently by two authors, and upon completion was reviewed by two researchers, with a third reviewer consulted in case of disagreement.

2.6. Data analysis and synthesis

The study was conducted using RevMan 5.3.5 software and Stata 17.0 software for data analysis, and continuous outcomes were expressed as standardized mean difference (SMD) with 95 % confidence interval (CI). Heterogeneity was tested by p values and I^2. Heterogeneity was considered low enough for meta-analysis using a fixed-effects model if $p > 0.10$ and $I^2 \langle 50$ %. If $p < 0.10$ and $I^2 \rangle 50$ %, heterogeneity was high, and a random effects model was used. Sensitivity analyses were performed for individual studies using a case-by-case exclusion method. If heterogeneity was excessive ($I^2 > 85$ %), meta-regression analysis was performed. The included studies were tested for bias using funnel plots and Egger's test.

3. Results

3.1. Search results

The search and screening process is shown in Fig. 1. A total of 1983 studies were retrieved from the database. After excluding duplicates, 929 studies remained. A total of 366 articles were excluded for not being RCTs, 33 articles were excluded due to missing data, and 532 articles were excluded because the subjects did not match. Additionally, 28 articles were drawn from the references, of which 11 articles were excluded for non-empirical literature, 5 articles were excluded due to missing data, and 9 articles were excluded because the subjects did not match. In the end, a total of 10 articles were included in the review and all were included in the meta-analysis.

3.2. Characteristics of the included studies

A total of 10 studies were included in this meta-analysis (Benzing and Schmidt, 2019; Dovis et al., 2015; Ji et al., 2023; Kollins et al., 2020; Lim et al., 2012, 2019, 2023; Medina et al., 2021; Qian et al., 2018), including 820 children with ADHD. The publication years of these studies ranged from 2012 to 2023, and the age range of the subjects was 6–12 years. The basic characteristics, experimental results, experimental design, and measurement methods of the included RCTs are shown in Table 1.

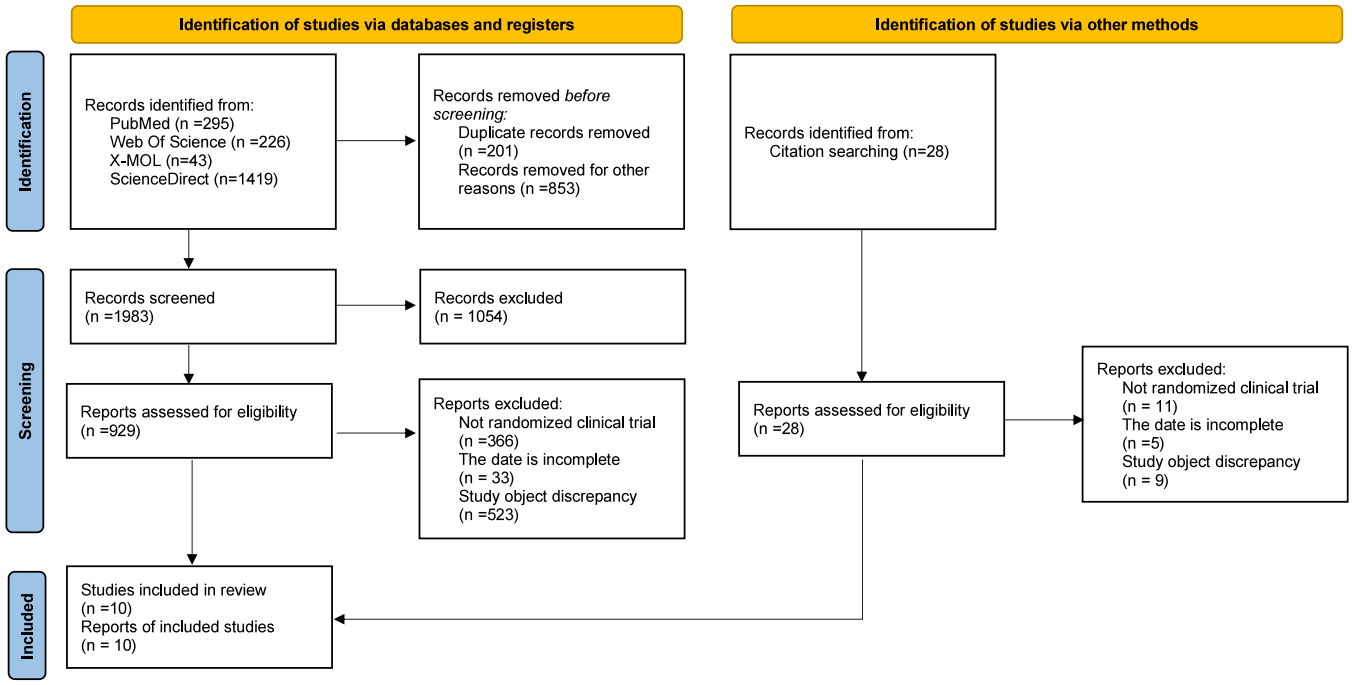


Fig. 1. Flow diagram of the literature search.

Table 1
Characteristics of studies included in the meta-analysis.

author	nations	Year of publication	age	sex	Intervention tools	frequency	experimental group	control subjects	gauge
HongQing Ji	Sino	2023	8–12 years	76 % male	ExerHeart	8 weeks 3 days/week 50min/day	*n* = 16 M = 262.44 sd=98.58	*n* = 14 M = 294.7 sd=103.07	FAIR scale
Scott H Kollins	United States of America	2020	8–12 years	69 % male	AKL-T01	4 weeks 5 days/week 25min/day	*n* = 169 M = 18.3 sd=3.1	*n* = 160 M = 18.3 sd=3.3	TOVA API
James F. Leckman	United States of America	2018	6–13 years	70 % men	ACTIVAT	8 weeks 6 times/week	*n* = 35 M = 15.4 sd=5.33	*n* = 35 M = 15.66 sd=4.88	ADHD-RS-IV
Choon Guan Lim	Singaporean	2012	6–12 years	80 % men	BCI	8 weeks 6 times/week	*n* = 81 M = 14.3 sd=5.68	*n* = 82 M = 14.08 sd=5.09	DSM-IV-TR scale
Choon Guan Lim	Singaporean	2019	6–12 years	85.5 % men	BCI	12 weeks 3 times/week	*n* = 15 M = 9.26 sd=2.78	*n* = 14 M = 9.14 sd=3.57	DSM-IV-TR scale
Refael Medina	Spanish	2021	8–11 years	28 males	KAD_SCI_01	12 weeks 3 times/week 15–20min/time	*n* = 28 M = 62.11 sd=5.76	*n* = 23 M = 62.25 sd=5.39	EDAH questionnaire
Sebastiaan Dovis	the Netherlands	2015	8–12 years	80 % men	Braining Brian	8 weeks 3 times/week 30min/times	*n* = 31 M = 11.9 sd=5.7	*n* = 31 M = 14 sd=6.3	DSN-IV scale
Mirko Schmidt	Switzerland	2019	8–12 years	82.4 % boys	Braining Brian	8 weeks 3 times/week 30min/time	*n* = 28 M = 62.11 sd=5.76	*n* = 23 M = 62.25 sd=5.39	DSM-IV-TR scale
Choon Guan Lim	Singaporean	2023	6–12 years	80 % boys	BCI	8 weeks 3 times/week	*n* = 10 M = 14.7 sd=4.97	*n* = 10 M = 14 sd=6.3	DSM-IV-TR scale
Xing Qian	Singaporean	2018	SD=9	100 % male	BCI	8 weeks 3 times/week	*n* = 18 M = 13.17 sd=4.08	*n* = 11 M = 17.28 sd=5.8	FAIR scale

3.3. Risk of bias

Six of the included studies described the method of randomized allocation, while the remaining studies referred to "randomization" without providing specific details. Five studies concealed the allocation scheme, while the others could have described it in detail. Four studies employed blinding, with the rest lacking specific descriptions of blinding procedures. Four studies reported outcome data comprehensively; two had missing data, and the others needed to provide detailed information. All ten studies reported primary and secondary outcomes, as illustrated in Figs. 2 and 3.

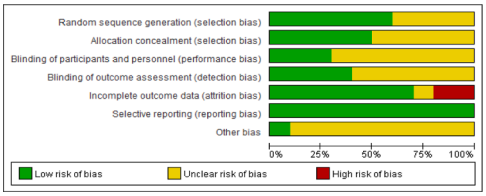


Fig. 2. Risk of bias graph.

3.4. Results of meta-analysis

A meta-analysis was conducted on the included studies, and the results are illustrated in Fig. 4-A. The test for heterogeneity among the included studies yielded X^2=9.97, $P = 0.35$, and I^2=10 %, indicating low heterogeneity. The combined effect size of the standardized mean difference (SMD) was −0.19, with a confidence interval (95 % CI) ranging from −0.34 to −0.03. The test for the combined effect size showed $Z = 2.40$, $p = 0.02$.

The analysis results indicated homogeneity among the included studies ($X^2 = 9.97$, $P = 0.35$, $I^2 = 10$ %). Therefore, a fixed-effects model was used to determine the combined effect sizes. The combined effect size for the standardized mean difference was −0.18, with confidence interval (95 % CI) ranging from −0.31 to −0.04 (Fig. 4-B)

These results suggest that video game interventions effectively improve attention in children with ADHD.

3.5. Bias test

The funnel plot of this study, shown in Fig. 5, appears to be symmetrical, indicating minimal bias.

The Egger test was used to assess bias in the included articles quantitatively, and the *p*-value of the Egger test was 0.13, which was greater than 0.05, indicating no publication bias in this study.

4. Discussion

Results suggest that videogame interventions can provide a statistically significant but not clinically meaningful improvement in the attention of children with ADHD. These findings validate previous studies and provide a reference for using video games to enhance attention in children with ADHD.

In the study, a human-computer interaction motor game was designed, and an 8-week intervention was conducted three times a week for 30 min per session (Benzing and Schmidt, 2019). The mean difference between the baseline and endpoint data of the experimental and control groups was 91.11 and 80.34, respectively, demonstrating an improvement in attention in children with ADHD. Similarly, In addition, there are also study used a motion-based video game, the mean differences between the baseline and endpoint data of the experimental and control groups were 5.6 and 4.1, respectively, showing the effectiveness of the video game intervention (Dovis et al., 2015).

In the study, an 8-week experiment using the ACTIVAT video game was conducted. The mean difference between the baseline and endpoint data for the experimental and control groups was 2.85 and 0.76, respectively, indicating a positive effect of the video game intervention (Bikic et al., 2018). Conversely, in the study, the AKL-T01 video game was used in a 4-week, 5-day-a-week experiment. The results showed that the video game did not significantly improve attention in children with ADHD (Kollins et al., 2020). This suggests that the duration of the intervention may affect the improvement in attention, which could be explored in future studies.

Additionally, analyzing the intervention designs used in the studies included in the meta-analysis revealed that video games combined with exercise had a more significant improvement in attention for children with ADHD compared to video game-only interventions. Previous studies have shown that physical activity alone can improve attention in children with ADHD (Kang et al., 2011). Therefore, combining video games with exercise may yield a more substantial improvement in attention for children with ADHD.

In this study, the effect size of video game intervention was 0.19, which was lower than that of stimulant drug intervention (0.5 - 0.8) and non-stimulant drug intervention (0.6 - 0.9) (Catalá-López et al., 2017). This indicates that video game intervention may not be a first-line intervention method, but it can be used as a complementary approach to drug intervention. Additionally, improvements in the video game intervention approach may be beneficial for enhancing the intervention effect and increasing the effect size.

Furthermore, while video game interventions have been associated with negative effects such as game addiction, exercise can bidirectionally modulate dopamine and its receptors. This neurotransmitter is associated with pleasure and reward mechanisms and has a significant impact on game addiction intervention (Ma and Liang, 2022).

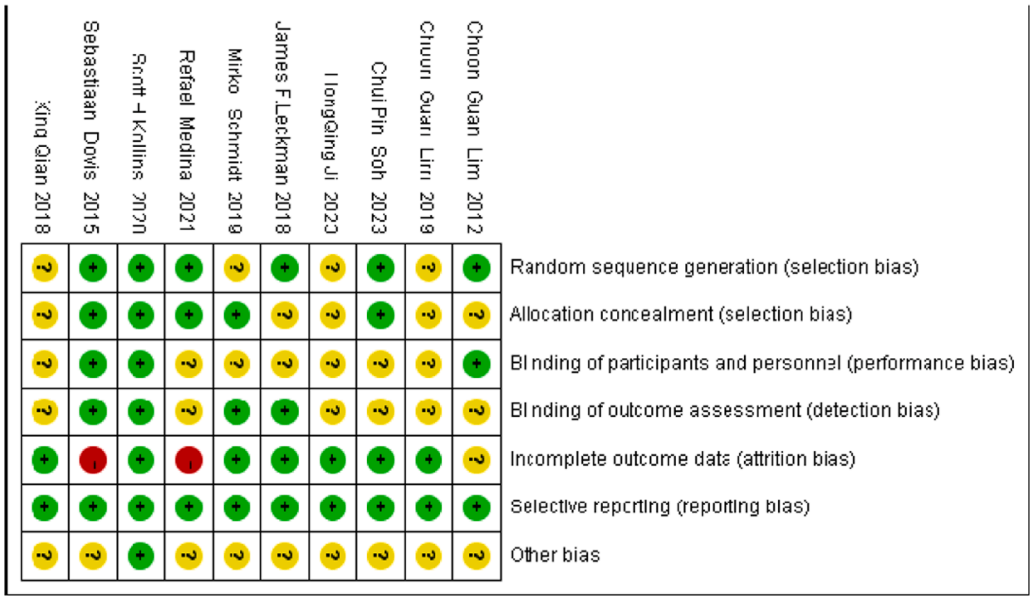


Fig. 3. Risk of bias summary.

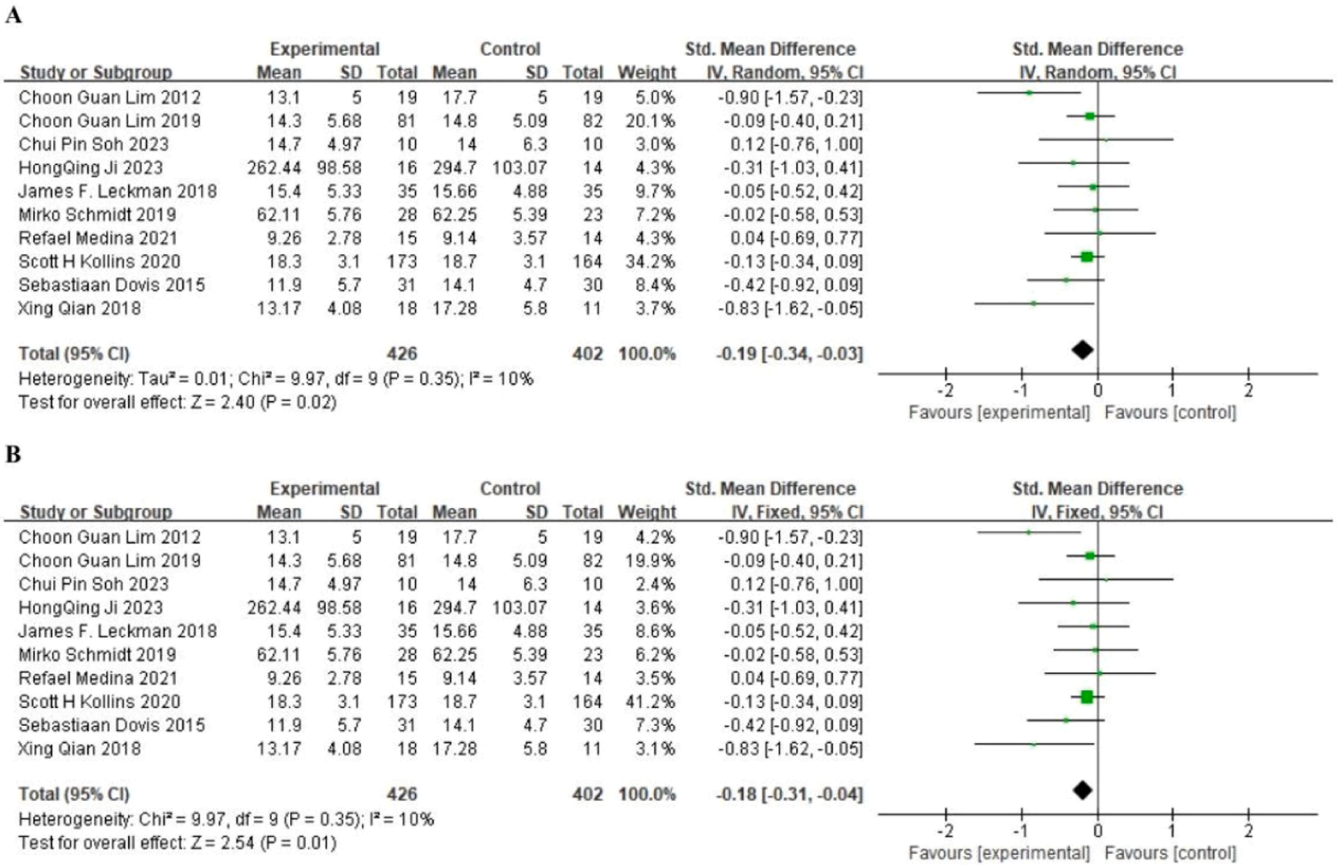


Fig. 4. Forest plots of effect size for attention deficit symptoms.

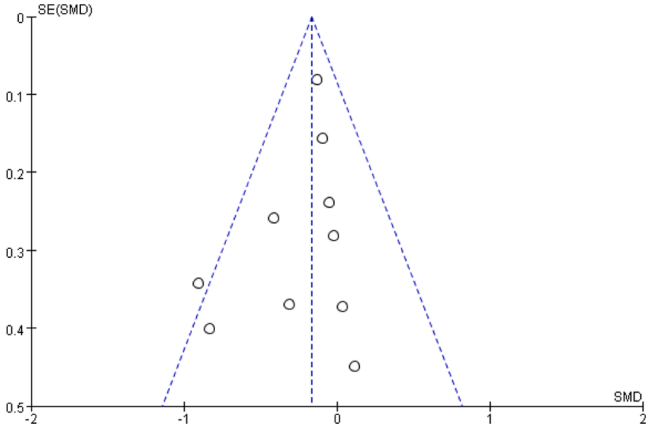


Fig. 5. Funnel plot for attention deficit symptoms.

Therefore, future studies on video game interventions for children with ADHD should consider incorporating exercise to enhance the effectiveness of the intervention and mitigate the negative effects of game addiction.

Even though this study suggests the benefits of videogame interventions, certain limitations of this report should be kept in mind. The authenticity and reliability of the results of this meta-analysis may be affected by several factors: (1) The studies included in this meta-analysis were published only in Chinese and English, which introduces a language bias. (2) Few studies specifically described the concealment of random assignment and blinding of subjects, researchers, and outcome assessors. (3) The lack of large-sample, multicenter, and long-term RCTs may affect the reliability of this meta-analysis.

5. Conclusion

Although the improvement from these interventions is statistically significant, it has not yet reached the level of clinical treatment efficacy, suggesting that video game interventions may only produce therapeutic effects when used in conjunction with other evidence-based therapies. This systematic review and meta-analysis provide supporting evidence for the feasibility of future video game intervention programs and suggest that future studies should focus on the intervention intensity, type, and optimal dose when combined with physical exercise.

CRediT authorship contribution statement

Pengwei Ma: Writing – original draft, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Zhuolin Xue:** Data curation. **Kun Yuan:** Validation. **Peiyun Zheng:** Validation. **Junfeng Li:** Validation, Methodology. **Jindong Chang:** Writing – review & editing, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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